

Pharmacy Sales Prediction Dataset Report from Onkod drug company in wanlweyn.

1. Introduction

This dataset was developed to support a machine learning project that predicts the number of pharmacy product sales for the following month. The objective is to train a supervised machine learning model using historical sales information so that it can estimate future sales accurately.

2. Dataset Objective

The purpose of the dataset is to:

- Train a machine learning model to forecast next month's sales.
- Analyze historical sales patterns.
- Improve inventory management.
- Reduce stock shortages and overstocking.
- Support business decision-making in pharmacy management.

3. Dataset Description

The dataset consists of **500 records**. Each record represents the sales information of a pharmacy product for a particular month.

The dataset contains **5 input features (X)** and **1 target variable (Y)**.

Features (X)

Feature	Description	Data Type
Month	Month number (1–12) representing the sales period	Integer
Opening_Stock	Quantity of stock available at the beginning of the month	Integer
Units_Sold_Current_Month	Number of units sold during the current month	Integer
Unit_Price_USD	Selling price of one unit in US Dollars	Decimal
Previous_Month_Sales	Number of units sold in the previous month	Integer

Target Variable (Y)

Target	Description
Next_Month_Sales	Number of units expected to be sold in the following month

4. Data Collection Method

The dataset was created using simulated pharmacy sales records that resemble real pharmacy operations. The values were generated to maintain realistic relationships between stock levels, sales, pricing, and future demand.

The dataset was cleaned before use by ensuring:

- No missing values.
 - Consistent numerical formats.
 - Appropriate data types.
 - Logical value ranges.
 - Ready for machine learning.
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5. Dataset Size

- Total Records: **500**
 - Input Features: **5**
 - Target Variable: **1**
 - File Format: **Microsoft Excel (.xlsx)**
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6. Machine Learning Problem

This project is a **Supervised Machine Learning Regression** problem because the objective is to predict a continuous numerical value (Next Month Sales).

Input variables (X) are used to predict the output variable (Y).

7. Possible Algorithms

The dataset can be used with the following machine learning algorithms:

- Linear Regression
 - Decision Tree Regression
 - Random Forest Regression
 - Gradient Boosting Regression
 - XGBoost Regression
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8. Expected Output

After training, the machine learning model receives only the five input features:

- Month
- Opening Stock
- Units Sold Current Month
- Unit Price
- Previous Month Sales

The model then predicts:

Next_Month_Sales

This prediction helps pharmacy managers estimate future demand and make better purchasing and inventory decisions.

9. Conclusion

The Pharmacy Sales Prediction Dataset provides a structured and clean dataset suitable for supervised machine learning. By learning from historical sales information, the trained model can predict future monthly sales, enabling pharmacies to improve inventory planning, reduce waste, avoid stock shortages, and make more informed business decisions.